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★ Chemical Reactions and Equations

★ Chemical Reactions

A process in which one or more substances react to form new substances (s) with new chemical identity and property.

Example :- Magnesium [Ribbon] + oxygen $\xrightarrow{\text{Heat}}$ Magnesium oxide [White powder]

- Some daily Life Examples are :-

1. Cooking of Food
2. Digestion of Food
3. Ripening of Fruit
4. Conversion of milk into curd.

Identification of Chemical Reactions

- We can easily identify if there is a chemical reaction occuring by seeing some Basic characteristics of a chemical reactions.

1. Evolution of Gas

We can Easily identify some of the chemical reaction by Evolution of gas, if there is Evolution of gas it means reaction is occuring

eg:- $Zn_{(s)} + H_2SO_{4(aq)} \longrightarrow ZnSO_{4(aq)} + H_{2(g)}$

2. Formation of Precipitate

- Some of the chemical Reactions are identified by the formation of Precipitate [ppt]

eg:- $Pb(NO_3)_{2(aq)} + KI_{(aq)} \longrightarrow PbI_{2(s)} + KNO_{3(aq)}$

Lead nitrate — Potassium iodide — Lead Iodide — Potassium Nitrate.

ii) $H_2SO_{4(aq)} + BaCl_{2(aq)} \longrightarrow BaSO_{4(s)} + 2HCl_{(aq)}$

Sulphuric acid — Barium chloride — Barium sulphate — Hydrochloric acid.

3. Change in Colour

eg:- $Fe + CuSO_4 \longrightarrow FeSO_4 + Cu$

Iron — copper sulphate (blue) — Iron sulphate (blue green) — Copper.

4. Change in Temperature

- When Quicklime [CaO] React with water to form slacked lime it releases a lot of energy in the form of Heat, this heat increases the temperature and we can identify that chemical reaction is occuring.

$$CaO + H_2O \longrightarrow Ca(OH)_2 + Heat$$

This type of reaction is known as Exothermic Reaction.

5. Change in state

- Reaction of Hydrogen with oxygen to form water.

$H_2(g) + O_2(g) \longrightarrow H_2O(l)$

gas + gas $\longrightarrow$ Liquid.

Chemical Equations

- A chemical equation is the symbolic representation of a chemical reaction. Symbols and formulae of the reactants and products are used for the same.

eg:- $CH_4(g) + 2O_2(g) \xrightarrow{\Delta} CO_2(g) + 2H_2O(l)$

Methane + oxygen $\longrightarrow$ Carbondioxide + Water

★ Things to remember

Physical state of the compound

s → Solid
g → gas
l → Liquid
aq → aqueous.

Balancing of Chemical Reactions

- There is a law which we have to follow while writing the chemical Reactions, that is Law of Conservation of Mass.

Law of Conservation of Mass

- According to Law of conservation of Mass: "Mass can neither be created nor be destroyed it can only be converted from one form to another form.

- It means that Mass is constant since the formation of universe and we are not creating it or destroying it we only convert it from one form to another and use it.

- This same Law we have to follow while writing down the chemical Reactions it means we have to take the same number of Atoms on the reactant and Product side of Each element.

$$2H_2 + O_2 \longrightarrow 2H_2O$$

This is a balanced chemical Reaction.

- A Balanced chemical Reaction has Equal number of Atoms of different Elements in the Reactants and Products.

- Steps used in balancing of chemical Reactions.

⇒ $Fe + H_2O \longrightarrow Fe_3O_4 + H_2$

step 1:- Check the number of atoms of each elements on the reactant and Products side of the equations.

Element	No. of atoms in Reactant side	No. of atoms in products side
Fe	1	3
H	2	2
O	1	4

(सबसे ज्यादा difference oxygen वाले case में हैं तो पहले oxygen को Balance करते हैं।)

step 2:- First of all, to balance oxygen atom we multiply H_2O molecule by 4.

$$Fe + 4(H_2O) \longrightarrow Fe_3O_4 + H_2$$

step 3:- Once again check the number of atoms of each elements on both sides.

Elements	LHS	RHS
Fe	1	3
H	8	2
O	4	4

(अब H में difference बड़ा आया।)

Step 4: Balancing Hydrogen atom by multiplying H_2 with 4.

$$Fe + 4(H_2O) \longrightarrow Fe_3O_4 + 4H_2$$

Now,

Element	LHS	RHS
Fe	1	3
H	8	8
O	4	4

Step 5: Balancing Fe by multiplying it by 3 on LHS

$$3(Fe) + 4(H_2O) \longrightarrow Fe_3O_4 + 4(H_2)$$

Now, We can see

Element	LHS	RHS
Fe	3	3
H	8	8
O	4	4

LHS = RHS

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Q:- Some equations you should Practice balancing (most likely equations)

i) $ZnCO_3 \longrightarrow ZnO + CO_2$

ii) $Pb(NO_3)_2 + KI \longrightarrow KNO_3 + PbI_2$

iii) $CO(g) + O_2(g) \longrightarrow CO_2(g)$

iv) $FeSO_4 \longrightarrow Fe_2O_3 + SO_2 + SO_3$

v) $AL + H_2SO_4 \longrightarrow Al_2(SO_4)_3 + H_2$

vi) $Zn + HNO_3 \longrightarrow Zn(NO_3)_2 + H_2O + N_2O$

Answers:

i) Already Balanced

ii) $Pb(NO_3)_2 + 2KI \longrightarrow 2KNO_3 + PbI_2$

iii) $2CO(g) + O_2(g) \longrightarrow 2CO_2(g)$

iv) $2FeSO_4 \longrightarrow Fe_2O_3 + SO_2 + SO_3$

v) $2AL + 3H_2SO_4 \longrightarrow Al_2(SO_4)_3 + 3H_2$

vi) $4Zn + 10HNO_3 \longrightarrow 4Zn(NO_3)_2 + 5H_2O + N_2O$

Types of Chemical Reactions

1:- Combination Reaction:-

This is the type of chemical Reactions in which two OR more than two Elements combine to form a single compound.

eg:- i) Quick Lime reacts with water to form slaked lime.

$$CaO(s) + H_2O(l) \longrightarrow Ca(OH)_2(aq) + Heat \downarrow$$

Calcium oxide (quick lime) — Calcium hydroxide (slaked lime) — exothermic

Imp:
⇒ Solution of calcium hydroxide (slaked lime) is used For white washing walls. Calcium hydroxide reacts slowly with CO_2 in air to form a thin layer of calcium carbonate, on the walls which gives a shiny appearance to the walls. Finally after 2 to 3 days of white washing calcium carbonate is formed.

$$Ca(OH)_2(aq) + CO_2(g) \longrightarrow CaCO_3(s) + H_2O(l)$$

Calcium hydroxide (slaked lime) — Calcium carbonate

ii) Formation of water from hydrogen gas & oxygen gas:

$$2H_2(g) + O_2(g) \longrightarrow 2H_2O(l)$$

Hydrogen oxygen water

iii) Burning of coal:

$$C(s) + O_2(g) \longrightarrow CO_2(g)$$

carbon oxygen carbon dioxide.

#2. Decomposition Reaction:

In a decomposition reaction, a single reactant breaks down into two or more simpler products. basically opposite of combination reaction.

Decomposition reaction are of 3 types:

i) Thermal Decomposition

These reactions use the energy in the form of heat for decomposition of the reactant.

eg:- a) Calcium carbonate on heating, decomposes to give calcium oxide and carbon dioxide.

$$CaCO_3(s) \xrightarrow{\Delta} CaO(s) + CO_2(g)$$

Calcium carbonate (Limestone) — Calcium oxide (Quick Lime)

★ Calcium oxide is used for manufacturing of cement.

b. Ferrous sulphate, the green colour crystals $FeSO_4 \cdot 7H_2O$ on heating lose water of crystallisation and forms dehydrated $FeSO_4$, which on decomposition gives ferric oxide, sulphur dioxide SO_2 and sulphur trioxide SO_3. Ferric oxide is a solid, while SO_2 and SO_3 are gases.

$$2FeSO_4(s) \xrightarrow{Heat} Fe_2O_3(s) + SO_2(g) + SO_3(g)$$

Ferrous sulphate → Ferric oxide, sulphur dioxide, sulphur trioxide

c. On heating lead nitrate, it decomposes to give yellow lead monoxide, nitrogen-dioxide and oxygen gas.

$$2Pb(NO_3)_2(s) \xrightarrow{Heat} 2PbO(s) + 4NO_2(g) + O_2(g)$$

Lead Nitrate (colourless) → Lead oxide (yellow), Nitrogen-dioxide (Brown Fumes), oxygen

(ii) Electrolytic Decomposition or Electrolysis

These reactions involve the use of electrical energy for the decomposition of the reactant molecules.

eg:- (a.) When electric current is passed through water, it decomposes to give oxygen and hydrogen.

$$2H_2O(l) \xrightarrow{\text{Electric current}} 2H_2(g) + O_2(g)$$

Water

(b) When electric current is passed through molten sodium chloride, it decomposes to give sodium metal and chlorine gas.

$$2NaCl(l) \xrightarrow{\text{Electric current}} 2Na(s) + Cl_2(g)$$

Molten sodium chloride

(iii) Photo Decomposition or Photolysis or Photochemical decomposition

These reactions involve the use of light energy for the purpose of decomposition.

Eg:- (a) When silver chloride is exposed to sunlight, it decomposes to give silver metal and chlorine gas.

$$2AgCl(s) \xrightarrow{\text{Sunlight}} 2Ag(s) + Cl_2(g)$$

Silver chloride (White) — Silver (Greyish white) — Chlorine (yellowish green)

(b) Similarly, silver bromide gives silver metal and bromine gas in the presence of sunlight.

$$2AgBr(s) \xrightarrow{\text{Sunlight}} Ag(s) + Br_2(g)$$

★ These reactions of silver Halides are used in black and White photography.

#3) Displacement Reaction :

These are those type of chemical Reactions in which more reactive elements displaces a less reactive element from its compound.

i) Single displacement Reaction :

It is a type of chemical reaction where an element reacts with a compound and takes the place of another element in that compound is called single displacement.

eg:- a) Zinc being more reactive the Cu, displaces Cu From $CuSO_4$ solution and forms new product, zinc sulphate and Cu metal.

$$Zn(s) + CuSO_4(aq) \longrightarrow ZnSO_4(aq) + Cu(s)$$

Copper sulphate solution (blue) → Zinc sulphate solution (colourless)

b) Similarly, Iron being more reactive than Cu displaces copper From aqueous solution of copper sulphate.

$$Fe(s) + CuSO_4(aq) \longrightarrow FeSO_4(aq) + Cu(s)$$

Copper sulphate (blue) → Ferrous sulphate or, Iron(II) sulphate (Green)

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ii) Double displacement Reaction :

The reaction in which two different atoms or group of atoms in the reactant molecules are displaced by each other is called double displacement reaction. it is also called Precipitation reaction as precipitate is also produced in such reactions.

Eg:- (a) On adding sodium sulphate to barium chloride, a curdy white precipitate of barium sulphate and a solution of sodium chloride are formed.

$$Na_2SO_4(aq) + BaCl_2(aq) \longrightarrow BaSO_4\downarrow + 2NaCl(aq)$$

Sodium sulphate — Barium chloride — Barium sulphate (White ppt) — Sodium chloride

(b) On adding silver nitrate solution to sodium bromide, a yellow precipitate of silver bromide and solution of sodium nitrate are formed.

$$AgNO_3(aq) + NaBr(aq) \longrightarrow AgBr\downarrow + NaNO_3(aq)$$

Silver Nitrate — Sodium Bromide — Silver Bromide (Yellow ppt) — Sodium Nitrate

#4: Oxidation Reaction:

A substance (in reactant side) is said to be oxidised if oxygen is added OR Hydrogen is removed to it after the reaction and this process of addition of oxygen and/or removal of Hydrogen is called oxidation.

eg:- $Cu + O_2 \longrightarrow CuO$

(Here Cu is oxidised due to addition of oxygen)

#5: Reduction:

A substance (in reactant side) is said to be reduced if oxygen is removed OR Hydrogen is added to it after the reaction and the process is called reduction.

eg:- $H_2 + F_2 \longrightarrow HF$

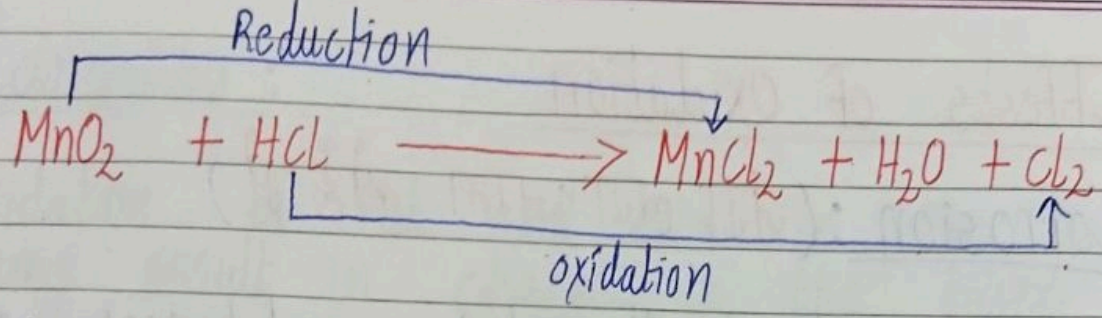
(Here F_2 is reduced due to addition of hydrogen)

#6: Redox Reaction:

It is a reaction in which oxidation and reduction both takes place simultaneously.
Also,
In a redox reaction substance getting oxidised is called Reducing agent.
And, the substance getting reduced is called oxidising agent.

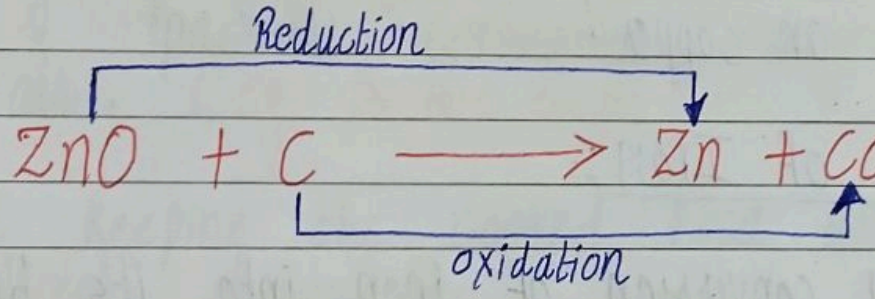
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Eg:-



Here,

MnO_2 is reduced to $MnCl_2$ (removal of oxygen)
HCl is oxidised to Cl_2 (removal of Hydrogen)
MnO_2 is oxidising agent
HCl is reducing agent



Here,

ZnO is reduced to Zn (removal of oxygen)
C is oxidised to CO (addition of oxygen)
ZnO is oxidising agent
C is reducing agent.

Effects of Oxidation

#1: Corrosion :(जंग लग जाना लोहे में)

When a metal is attacked by substances around it such as moisture, acid, air, water etc. It is said to be corrode & this process is called corrosion.

eg: Rusting of metal, blackening of silver, green coating on copper.

Rusting of Iron:-

The slow conversion of iron into its hydrated oxide, in the presence of moisture and air is called rusting, whereas the hydrated oxide of iron is called rust.

$$4Fe + 3O_2 \longrightarrow 2Fe_2O_3$$

Iron oxygen Ferric oxide

$$Fe_2O_3 + xH_2O \longrightarrow Fe_2O_3 \cdot xH_2O$$

Ferric oxide water Hydrated Ferric oxide

The brownish residue ($Fe_2O_3 \cdot xH_2O$) is commonly called rust and the phenomenon is called rusting.

Rancidity :

Oxidation of Fat and oils in the Food items result in change in the taste and smell. We say that Food items are rancid and the process is called rancidity.

Methods to prevent Rancidity :

★ Packing of Food items Like Potato Wafers etc in packets containing nitrogen gas instead of air. (eg : packed Chips)

★ Avoid keeping the cooked Food and Food materials in direct sunlight.

★ Keeping Food materials in air-tight containers.

★ Refrigeration of cooked Food.

★ Adding antioxidants Like Butylated hydroxyanisole to food items.